

**E. Jeffrey Samuels (192524412)**

## **SET 9 – ONLINE LEARNING ACTIVITY ANALYSIS**

### **Aim**

To analyze online learning activities and student performance using histograms, boxplots, scatter plots, and line charts in R.

### **Algorithm**

1. Import the online learning dataset.
2. Convert Login\_Date into Date format.
3. Create a histogram of Quiz\_Score.
4. Create a boxplot of Quiz\_Score by Course.
5. Create a scatter plot between Study\_Time and Quiz\_Score.
6. Represent Videos\_Watched using point size and transparency.
7. Calculate the average quiz score for each month.
8. Create a line chart to identify the performance trend.
9. Display the dataset.

### **R Program**

```
library(ggplot2)
```

```
data <- data.frame(  
  Student_ID=c("L01","L02","L03","L04","L05","L06"),  
  Gender=c("Male","Female","Male","Female","Male","Female"),  
  Age=c(20,22,19,21,23,20),  
  Course=c("R","R","SQL","R","R","SQL"),  
  Study_Time=c(3.5,4.2,2.0,5.0,2.5,4.0),  
  Videos_Watched=c(12,15,8,18,9,14),  
  Quiz_Score=c(78,85,65,92,70,88),  
  Login_Date=as.Date(c("2025-01-05","2025-01-05","2025-02-08",  
                        "2025-02-08","2025-03-12","2025-03-12"))  
)
```

```
ggplot(data, aes(Quiz_Score)) +
```

```
geom_histogram(binwidth=5) +  
labs(title="Quiz Score Distribution", x="Quiz Score", y="Frequency")
```

```
ggplot(data, aes(Course, Quiz_Score)) +  
  geom_boxplot() +  
  labs(title="Quiz Score by Course", x="Course", y="Quiz Score")
```

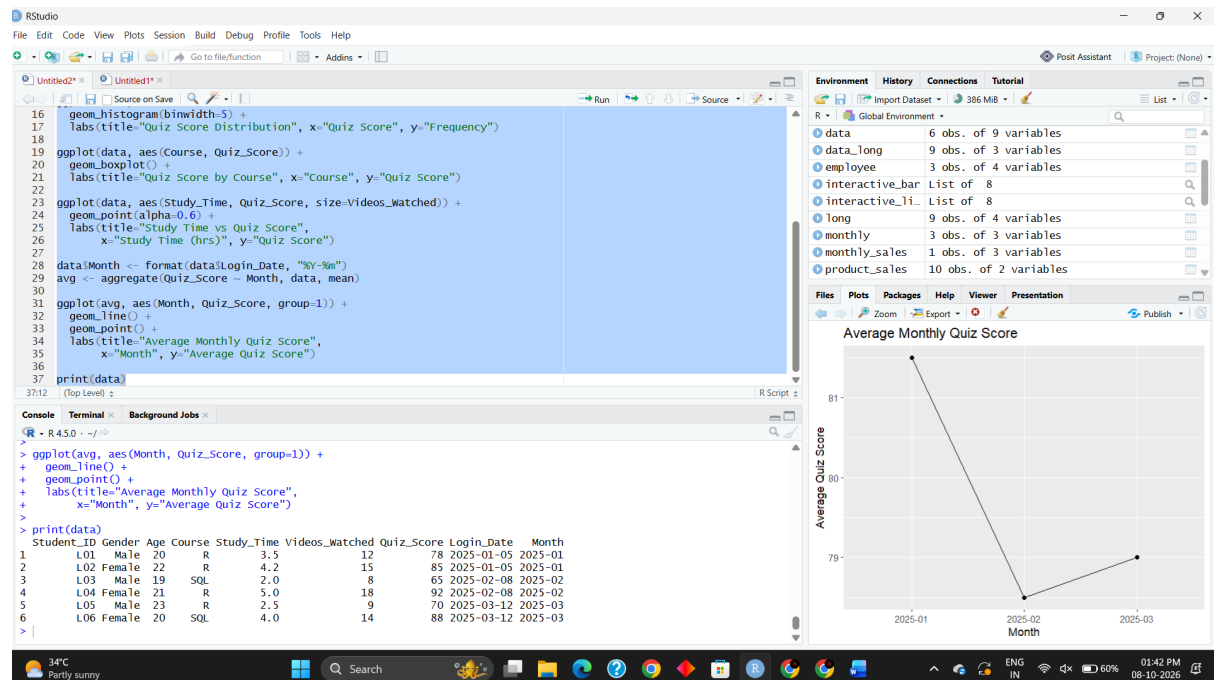
```
ggplot(data, aes(Study_Time, Quiz_Score, size=Videos_Watched)) +  
  geom_point(alpha=0.6) +  
  labs(title="Study Time vs Quiz Score",  
        x="Study Time (hrs)", y="Quiz Score")
```

```
data$Month <- format(data$Login_Date, "%Y-%m")  
avg <- aggregate(Quiz_Score ~ Month, data, mean)
```

```
ggplot(avg, aes(Month, Quiz_Score, group=1)) +  
  geom_line() +  
  geom_point() +  
  labs(title="Average Monthly Quiz Score",  
        x="Month", y="Average Quiz Score")
```

```
print(data)
```

## OUTPUT:



## Result

The histogram shows the distribution of quiz scores, while the boxplot compares quiz performance between R and SQL courses. The scatter plot shows the relationship between study time, quiz score, and videos watched. The line chart displays the average monthly quiz score trend successfully.